

Record 4 of 10



The U.S. government does not review or approve the safety and science of all studies listed on this website.

Read our full [disclaimer](https://clinicaltrials.gov/about-site/disclaimer) (<https://clinicaltrials.gov/about-site/disclaimer>) for details.

Phase 2a, Exploratory Study to Evaluate the Safety, Efficacy, Tolerability and Pharmacokinetics of XPF-002 in Patients With Primary/Inherited Erythromelalgia

Completed ⓘ

No longer looking for participants

ClinicalTrials.gov ID ⓘ NCT01486446

Sponsor ⓘ Xenon Pharmaceuticals Inc.

Information provided by ⓘ Xenon Pharmaceuticals Inc. (Responsible Party)

Last Update Posted ⓘ 2014-04-14

Results Posted Tab

Results Overview

Conditions ⓘ

Primary Erythromelalgia

Inherited Erythromelalgia

Intervention/Treatment ⓘ

- Drug: XPF-002
- Drug: Placebo

Other Study ID Numbers ⓘ

- XPF-002-202

Study Design

Allocation ⓘ: Randomized

Interventional Model ⓘ: Parallel Assignment

Masking ⓘ: Quadruple (Participant, Care Provider, Investigator, Outcomes Assessor)

Primary Purpose ⓘ: Treatment

Results Point of Contact

Name/Title: Dr. Y Paul Goldberg

Organization: Xenon Pharmaceuticals Inc.

Phone: 604 484 3300

Email: pgoldberg@xenon-pharma.com

Enrollment (Actual) ⓘ

8

Study Type ⓘ

Interventional

Study Record Dates

These dates track the progress of study record and summary results submissions to ClinicalTrials.gov. Study records and reported results are reviewed by the National Library of Medicine (NLM) to make sure they meet specific quality control standards before being posted on the public website.

Study Registration Dates	Results Reporting Dates	Study Record Updates
First Submitted ⓘ 2011-12-02 First Submitted that Met QC Criteria ⓘ 2011-12-02 First Posted (Estimated) ⓘ 2011-12-06	Results First Submitted ⓘ 2014-01-21 Results First Submitted that Met QC Criteria ⓘ 2014-03-10 Results First Posted (Estimated) ⓘ 2014-04-14	Last Update Submitted that Met QC Criteria ⓘ 2014-03-10 Last Update Posted (Estimated) ⓘ 2014-04-14 Last Verified ⓘ 2014-03

Participant Flow ⓘ

Recruitment Details	[Not Specified]
Pre-assignment Details	[Not Specified]

Arm/Group Title	XPF-002	Placebo
Arm/Group Description	XEN402 8% w/w ointment for topical application, applied to the skin twice daily for 14 or 21 days	XEN402 0% w/w ointment for topical application. Identical content and appearance to the XPF-002 ointment but without the active medicine. Applied to the skin twice daily for 14 or 21 days.

Period Title: **Overall Study**

Arm/Group Title	XPF-002	Placebo
Arm/Group Description	XEN402 8% w/w ointment for topical application, applied to the skin twice daily for 14 or 21 days	XEN402 0% w/w ointment for topical application. Identical content and appearance to the XPF-002 ointment but without the active medicine. Applied to the skin twice daily for 14 or 21 days.
Started	7	1
Completed	6	1
Not Completed	1	0
Reason Not Completed		
Adverse Event	1	0

Baseline Characteristics

Arm/Group Title	XPF-002	Placebo	Total
Arm/Group Description	XEN402 8% w/w ointment for topical application, applied to the skin twice daily for 14 or 21 days	XEN402 0% w/w ointment for topical application. Identical content and appearance to the XPF-002 ointment but without the active medicine. Applied to the skin twice daily for 14 or 21 days.	Total of all reporting groups
Overall Number of Baseline Participants	7	1	8
Baseline Analysis Population Description	[Not Specified]		

[Expand](#)

Age, Categorical

Measure Type: Count of Participants | Unit of measure: Participants

Number Analyzed	7 participants	1 participants	8 participants
<=18 years	00.0%	00.0%	00.0%
Between 18 and 65 years	685.7%	00.0%	675.0%
>=65 years	114.3%	1100.0%	225.0%

Age, Continuous

Mean (Standard Deviation) | Unit of measure: years

Number Analyzed	7 participants	1 participants	8 participants
	39.7 (19.3)	77 ^[1] (NA)	44.4 (22.2)

[1] Only 1 placebo subject, therefore no SD.

Arm/Group Title	XPF-002	Placebo	Total
Description	topical application, applied to the skin twice daily for 14 or 21 days	topical application. Identical content and appearance to the XPF-002 ointment but without the active medicine. Applied to the skin twice daily for 14 or 21 days.	
Overall Number of Baseline Participants	7	1	8
Baseline Analysis Population Description	[Not Specified]		

Number Analyzed	7 participants	1 participants	8 participants
Female	6 85.7%	1 100.0%	7 87.5%
Male	1 14.3%	0 0.0%	1 12.5%

Region of Enrollment

Measure Type: Number | Unit of measure: participants

Number Analyzed	7 participants	1 participants	8 participants
United States	7	1	8

Outcome Measures

[Expand all](#) / [Collapse all](#)

1. Average Daily Use of Cooling for Erythromelalgia-Related Pain in Treatment Period 2

Type: Primary | Time Frame: 14-21 Days

Description	Using diary cards, subjects recorded the use of all non-pharmacological cooling methods used to relieve their erythromelalgia (EM) pain each day during Treatment Period 2. A smaller average number of cooling uses each day in Treatment Period 2 indicates that subjects were in less pain/required less use of cooling to relieve their EM pain and vice versa.
Time Frame	14-21 Days
Analysis Population Description	[Not Specified]

Arm/Group Title	XPF-002	Placebo
Arm/Group Description	XEN402 8% w/w ointment for topical application, applied to the skin twice daily for 14 or 21 days	XEN402 0% w/w ointment for topical application. Identical content and appearance to the XPF-002 ointment but without the active medicine. Applied to the skin twice daily for 14 or 21 days.
Overall Number of Participants Analyzed	7	1
Mean (Inter-Quartile Range) Unit of Measure: cooling uses/day	0.24 (0.03 to 0.27)	2.4 ^[1] (NA to NA)
[1] Only 1 placebo subject, therefore no IQR.		

2. Average Cooling Duration (Minutes Per Day) for EM-related Pain in Treatment Period 2

Type: Other Pre-specified | Time Frame: 14-21 Days

Description	Using diary cards, subjects recorded the use and duration of all non-pharmacological cooling methods used to relieve their EM pain each day during Treatment Period 2. A smaller average duration of cooling each day in Treatment Period 2 indicates that subjects were in less pain/required less use of cooling to relieve their EM pain and vice versa.	
Time Frame	14-21 Days	
Analysis Population Description	[Not Specified]	
Arm/Group Title	XPF-002	Placebo
Arm/Group Description	XEN402 8% w/w ointment for topical application, applied to the skin twice daily for 14 or 21 days	XEN402 0% w/w ointment for topical application. Identical content and appearance to the XPF-002 ointment but without the active medicine. Applied to the skin twice daily for 14 or 21 days.
Overall Number of Participants Analyzed	7	1
Mean (Inter-Quartile Range) Unit of Measure: minutes/day	7.8 (0 to 13.3)	86 ^[1] (NA to NA)
[1] Only 1 placebo subject, therefore no IQR.		

3. Time to Exit (Minutes) From Standard Heat Inductions in Treatment Period 1, Compared to Baseline

Type: Other Pre-specified | Time Frame: Baseline to Day 5

Description	A standard heat induction was performed on multiple occasions pre-and post treatment. Subjects placed their feet in front of an electric heater for up to 75 minutes. Subjects rated their pain intensity/severity using numerical and categorical rating scales at fixed intervals before, during and after the heating procedure. The heating procedure continued until one of the pre-defined stopping criteria were met, or when the 75 minutes were over. Subjects could stop the procedure at any time if pain was intolerable. A stopwatch was used to record the time the heating procedure was stopped. This time is known as "Time to Exit".	
-------------	---	--

	A longer Time to Exit compared to Baseline indicates that subjects were able to tolerate the heat for a longer period when on treatment.	
Time Frame	Baseline to Day 5	
Analysis Population Description	[Not Specified]	
Arm/Group Title	XPF-002	Placebo
Arm/Group Description	XEN402 8% w/w ointment for topical application, applied to the skin twice daily for 14 or 21 days	XEN402 0% w/w ointment for topical application. Identical content and appearance to the XPF-002 ointment but without the active medicine. Applied to the skin twice daily for 14 or 21 days.
Overall Number of Participants Analyzed	7	1
Mean (Inter-Quartile Range) Unit of Measure: Percentage of Baseline	138 (78 to 188)	96 ^[1] (NA to NA)
[1] Only 1 placebo subject, therefore no IQR.		

4. Time to Exit (Minutes) From Standard Heat Inductions in Treatment Period 2, Compared to Baseline

Type: Other Pre-specified | Time Frame: Baseline and 14 or 21 days

Description	A standard heat induction was performed on multiple occasions pre-and post treatment. Subjects placed their feet in front of an electric heater for up to 75 minutes. Subjects rated their pain intensity/severity using numerical and categorical rating scales at fixed intervals before, during and after the heating procedure. The heating procedure continued until one of the pre-defined stopping criteria were met, or when the 75 minutes were over. Subjects could stop the procedure at any time if pain was intolerable. A stopwatch was used to record the time the heating procedure was stopped. This time is known as "Time to Exit". A longer Time to Exit compared to Baseline indicates that subjects were able to tolerate the heat for a longer period when on treatment.	
Time Frame	Baseline and 14 or 21 days	
Analysis Population Description	[Not Specified]	
Arm/Group Title	XPF-002	Placebo
Arm/Group Description	XEN402 8% w/w ointment for topical application, applied to the skin twice daily for 14 or 21 days	XEN402 0% w/w ointment for topical application. Identical content and appearance to the XPF-002 ointment but without the active medicine. Applied to the skin twice daily for 14 or 21 days.
Overall Number of Participants Analyzed	7	1
Mean (Inter-Quartile Range) Unit of Measure: Percentage of Baseline	100 (92 to 112)	107 ^[1] (NA to NA)
[1] Only 1 placebo subject, therefore no IQR.		

5. Time to Moderate Pain (Minutes) During Standard Heat Inductions in Treatment Period 1, Compared to Baseline

Type: Other Pre-specified | Time Frame: Baseline to Day 5

Description	A standard heat induction was performed on multiple occasions pre-and post treatment. Subjects placed their feet in front of an electric heater for up to 75 minutes. Subjects rated their pain intensity/severity using numerical and categorical rating scales at fixed intervals before, during and after the heating procedure. The heating procedure continued until one of the pre-defined stopping criteria were met, or when the 75 minutes were over. Subjects could stop the procedure at any time if pain was intolerable. A stopwatch was used to record the time the subject recorded a pain intensity numerical score of 5 or more (on a scale of 0-10). This time is known as "Time to Moderate Pain". A longer Time to Moderate Pain compared to Baseline indicates that subjects were able to tolerate the heat for a longer period when on treatment.	
Time Frame	Baseline to Day 5	
Analysis Population Description	[Not Specified]	
Arm/Group Title	XPF-002	Placebo
Arm/Group Description	XEN402 8% w/w ointment for topical application, applied to the skin twice daily for 14 or 21 days	XEN402 0% w/w ointment for topical application. Identical content and appearance to the XPF-002 ointment but without the active medicine. Applied to the skin twice daily for 14 or 21 days.
Overall Number of Participants Analyzed	7	1
Mean (Inter-Quartile Range) Unit of Measure: Percentage of Baseline	137 (74 to 172)	90 ^[1] (NA to NA)
[1] Only 1 placebo subject, therefore no IQR.		

6. Time to Moderate Pain (Minutes) During Standard Heat Inductions in Treatment Period 2, Compared to Baseline

Type: Other Pre-specified | Time Frame: Baseline and 14 or 21 days

Description	A standard heat induction was performed on multiple occasions pre-and post treatment. Subjects placed their feet in front of an electric heater for up to 75 minutes. Subjects rated their pain intensity/severity using numerical and categorical rating scales at fixed intervals before, during and after the heating procedure. The heating procedure continued until one of the pre-defined stopping criteria were met, or when the 75 minutes were over. Subjects could stop the procedure at any time if pain was intolerable. A stopwatch was used to record the time the subject recorded a pain intensity numerical score of 5 or more (on a scale of 0-10). This time is known as "Time to Moderate Pain". A longer Time to Moderate Pain compared to Baseline indicates that subjects were able to tolerate the heat for a longer period when on treatment.	
Time Frame	Baseline and 14 or 21 days	
Analysis Population Description	[Not Specified]	

Arm/Group Title	XPF-002	Placebo
Arm/Group Description	XEN402 8% w/w ointment for topical application, applied to the skin twice daily for 14 or 21 days	XEN402 0% w/w ointment for topical application. Identical content and appearance to the XPF-002 ointment but without the active medicine. Applied to the skin twice daily for 14 or 21 days.
Overall Number of Participants Analyzed	7	1
Mean (Inter-Quartile Range) Unit of Measure: Percentage of Baseline	103 (91 to 115)	135 ^[1] (NA to NA)
[1] Only 1 placebo subject, therefore no IQR.		

7. Daily Pain (Maximum Pain Intensity) in Treatment Period 2, Compared to Baseline

Type: Other Pre-specified | Time Frame: Baseline to 14 or 21 days

Description	During Baseline and Treatment Period 2, subjects recorded pain scores in their diary cards 3 times each day (upon waking, lunchtime and evening). On each recording occasion, subjects recorded the maximum pain experienced since the previous recording occasion using an 11 point numerical rating scale of pain intensity, PINRS (where 0 = no pain and 10 = worst pain imaginable). A lower score compared to Baseline indicates that the subjects experienced less severe pain on treatment than during Baseline.	
Time Frame	Baseline to 14 or 21 days	
Analysis Population Description	[Not Specified]	
Arm/Group Title	XPF-002	Placebo
Arm/Group Description	XEN402 8% w/w ointment for topical application, applied to the skin twice daily for 14 or 21 days	XEN402 0% w/w ointment for topical application. Identical content and appearance to the XPF-002 ointment but without the active medicine. Applied to the skin twice daily for 14 or 21 days.
Overall Number of Participants Analyzed	7	1
Mean (Inter-Quartile Range) Unit of Measure: Percentage of Baseline	78.5 (75.9 to 94.2)	76 ^[1] (NA to NA)
[1] Only 1 placebo subject, therefore there is no IQR.		

8. Sleep Interference Due to Pain in Treatment Period 2, Compared to Baseline

Type: Other Pre-specified | Time Frame: Baseline to 14 or 21 days

Description	During Baseline and Treatment Period 2, subjects recorded sleep interference scores in their diary cards for each night (scores were recorded upon waking). Sleep Interference due to pain is scored using an 11 point numerical rating scale where 0 = pain does not interfere with sleep and 10 = completely interferes, unable to sleep due to pain.	
-------------	--	--

	A lower sleep interference score compared to Baseline indicates that the subjects' pain interfered with sleep less on treatment than during Baseline.	
Time Frame	Baseline to 14 or 21 days	
Analysis Population Description	[Not Specified]	
Arm/Group Title	XPF-002	Placebo
Arm/Group Description	XEN402 8% w/w ointment for topical application, applied to the skin twice daily for 14 or 21 days	XEN402 0% w/w ointment for topical application. Identical content and appearance to the XPF-002 ointment but without the active medicine. Applied to the skin twice daily for 14 or 21 days.
Overall Number of Participants Analyzed	7	1
Mean (Inter-Quartile Range) Unit of Measure: Percentage of Baseline	76.5 (48.8 to 86.9)	46 ^[1] (NA to NA)
[1] Only 1 placebo subject, therefore no IQR.		

Adverse Events

Time Frame	[Not Specified]
Adverse Event Reporting Description	[Not Specified]

Arm/Group Title	XPF-002	Placebo
Arm/Group Description	XEN402 8% w/w ointment for topical application, applied to the skin twice daily for 14 or 21 days	XEN402 0% w/w ointment for topical application. Identical content and appearance to the XPF-002 ointment but without the active medicine. Applied to the skin twice daily for 14 or 21 days.

[Expand](#)

All-Cause Mortality

Arm/Group Title	XPF-002	Placebo
	Affected / at Risk (%)	Affected / at Risk (%)
Total	--/--	--/--

Serious Adverse Events

Arm/Group Title	XPF-002		Placebo	
	Affected / at Risk (%)	# Events	Affected / at Risk (%)	# Events
Total	1/7 (14.29%)		0/1 (0.00%)	

Vascular disorders

Arm/Group Title	XPF-002		Placebo	
Arm/Group Description	XEN402 8% w/w ointment for topical application, applied to the skin twice daily for 14 or 21 days		XEN402 0% w/w ointment for topical application. Identical content and appearance to the XPF-002 ointment but without the active medicine. Applied to the skin twice daily for 14 or 21 days.	
Hospitalization due to worsening erythromelalgia pain ^[1]	1/7 (14.29%)	1	0/1 (0.00%)	0
[1] Subject was admitted to hospital for treatment of worsening erythromelalgia pain 131 days following their last dose of study drug.				

Other (Not Including Serious) Adverse Events

Frequency Threshold for Reporting Other Adverse Events	0%			
Arm/Group Title	XPF-002		Placebo	
	Affected / at Risk (%)	# Events	Affected / at Risk (%)	# Events
Total	7/7 (100.00%)		1/1 (100.00%)	

Ear and labyrinth disorders

External ear inflammation ¹	1/7 (14.29%)	2	0/1 (0.00%)	0
--	--------------	---	-------------	---

Eye disorders

Eye inflammation ¹	1/7 (14.29%)	1	0/1 (0.00%)	0
-------------------------------	--------------	---	-------------	---

Gastrointestinal disorders

Nausea ¹	2/7 (28.57%)	2	0/1 (0.00%)	0
Oopharyngeal pain ¹	1/7 (14.29%)	2	0/1 (0.00%)	0
Tooth infection ¹	1/7 (14.29%)	1	0/1 (0.00%)	0

General disorders

Nodule ¹	1/7 (14.29%)	4	0/1 (0.00%)	0
Flushing ¹	1/7 (14.29%)	1	0/1 (0.00%)	0
Malaise ¹	1/7 (14.29%)	1	0/1 (0.00%)	0
Pain ¹	0/7 (0.00%)	0	1/1 (100.00%)	1

Musculoskeletal and connective tissue disorders

Arthralgia ¹	1/7 (14.29%)	1	0/1 (0.00%)	0
Muscle spasms ¹	1/7 (14.29%)	1	0/1 (0.00%)	0

Nervous system disorders

Headache ¹	3/7 (42.86%)	8	0/1 (0.00%)	0
Paraesthesia ¹	2/7 (28.57%)	3	0/1 (0.00%)	0
Burning sensation ¹	1/7 (14.29%)	1	0/1 (0.00%)	0

Arm/Group Title	XPF-002		Placebo	
Arm/Group Description	XEN402 8% w/w ointment for topical application, applied to the skin twice daily for 14 or 21 days		XEN402 0% w/w ointment for topical application. Identical content and appearance to the XPF-002 ointment but without the active medicine. Applied to the skin twice daily for 14 or 21 days.	
Sinus headache ¹	1/7 (14.29%)	1	0/1 (0.00%)	0

Renal and urinary disorders

Urinary tract infection ¹	1/7 (14.29%)	1	0/1 (0.00%)	0
--------------------------------------	--------------	---	-------------	---

Respiratory, thoracic and mediastinal disorders

Nasopharyngitis ¹	3/7 (42.86%)	3	0/1 (0.00%)	0
Upper-airway cough syndrome ¹	1/7 (14.29%)	2	0/1 (0.00%)	0
Cough ¹	1/7 (14.29%)	1	0/1 (0.00%)	0
Rhinorrhoea ¹	1/7 (14.29%)	1	0/1 (0.00%)	0

Skin and subcutaneous tissue disorders

Rash ¹	2/7 (28.57%)	3	0/1 (0.00%)	0
Erythema ¹	1/7 (14.29%)	2	0/1 (0.00%)	0
Rash pruritic ¹	1/7 (14.29%)	1	0/1 (0.00%)	0
Skin exfoliation ¹	0/7 (0.00%)	0	1/1 (100.00%)	1

Vascular disorders

Flushing ¹	1/7 (14.29%)	3	0/1 (0.00%)	0
Cyanosis ¹	1/7 (14.29%)	1	0/1 (0.00%)	0

¹ Term from vocabulary, MedDRA 15.0

Limitations and Caveats

This was a small trial with many exploratory efficacy measures. There was no primary outcome measure, all measures are "other: pre-specified".

Only 8 subjects were enrolled (1 placebo), the actual number of Adverse Events is used, not incidence.

Collaborators and Investigators

This is where you will find people and organizations involved with this study.

Sponsor ⓘ

Xenon Pharmaceuticals Inc.

Investigators ⓘ

- Principal Investigator: Almena L Free, MD, Pinnacle Research Group LLC.

More Information

Record History

Certain Agreements ⓘ

Principal Investigators are NOT employed by the organization sponsoring the study.

There IS an agreement between Principal Investigators and the Sponsor (or its agents) that restricts the PI's rights to discuss or publish trial results after the trial is completed